A General Contest Inversion Algorithm

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Abstract

Multinomial probit choice probabilities are Gaussian orthant integrals, and for thirty years econometrics has computed them by simulation, most commonly with the GHK sampler, at cost $O(n^2R)$ per probability vector and error $O(R^{-1/2})$. We observe that the choice event is one-dimensional: the probability that alternative i wins is an integral over the winning value, and one shared survival field prices all n alternatives at once. For covariances under which performances are conditionally independent given a few shared variables—factor, block, nested and hierarchical structures, which are also the covariances multinomial probit can identify—the field method is deterministic, computes all n probabilities and their exact Jacobian in $O(nLQ)$ operations, and inverts observed probabilities back to utilities. At $n = 200$ it is two hundred times faster than GHK at 10^4 draws and five times more accurate, and the measured gap grows as $n^{2.8}$. The same two calls price and invert softmax, Thurstone–Mosteller, factor multinomial probit, and a hierarchy of team and sector models, to ten million alternatives.

1 Introduction

A decision maker faces n alternatives and chooses the one with the highest utility. Write $U_i = \mu_i + \varepsilon_i$ with $\varepsilon \sim N(0, \Sigma)$; the probability that alternative i is chosen is

$$p_i \; = \; \Pr\{U_i \geq U_j \text{ for all } j\} \; = \; \Pr\{\tilde{U} \geq 0\}, \qquad \tilde{U}_j = U_i - U_j, \tag{1}$$

an $(n-1)$-dimensional Gaussian orthant integral. This is the multinomial probit model. Its appeal has been understood since Thurstone (1927): unlike the logit family it imposes no independence of irrelevant alternatives, because Σ lets alternatives be close substitutes. Its curse has been understood just as long. The integral (1) has no closed form, and estimation needs it, and its derivatives, at every trial value of the parameters.

The workhorse answer is the GHK simulator of Geweke (1989), Börsch-Supan and Hajivassiliou (1993), Keane (1994) and Hajivassiliou et al. (1996). GHK factors Σ by Cholesky and writes the orthant probability as a nested sequence of one-dimensional truncated normal probabilities, each conditioning on draws for the previous coordinates. Averaging R such importance weights gives an unbiased, smooth estimate of one p_i at cost $O(n^2)$ per draw. The estimator made maximum simulated likelihood practical and it remains the default in textbooks (Train,

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2009). Its costs are also standard: error shrinks as $O(R^{-1/2})$; the log of a simulated probability is biased at $O(1/R)$; gradients inherit the simulation noise; and a full probability vector, which the likelihood needs, costs n separate runs.

This paper takes a different route to the same object. The choice event is one-dimensional. Alternative i wins when its performance beats the field, so

$$p_i \;=\; \int f_i(x) \prod_{j \neq i} S_j(x)\,dx, \tag{2}$$

where f_i is the density of U_i and S_j the survival of U_j: integrate over the winning value x, requiring everyone else to fall short. For independent performances Cotton (2021) computed (2) for all n on a lattice by building the field $\sum_j \log S_j$ once and dividing each alternative out, and inverted the map from probabilities back to μ. The present paper extends the field construction to correlated performances by conditioning: whenever Σ makes performances independent given a few shared variables, the field factorizes, and all n probabilities, their exact Jacobian, and the inversion cost $O(nLQ)$ for a lattice of L points and Q quadrature nodes. No simulation, no per-alternative repetition, deterministic error.

The structures that admit this treatment are not a restriction to apologize for. Unrestricted Σ is not identified in multinomial probit; applied work restricts to factor-analytic and related structured covariances as a matter of course (Train, 2009). The table lists the models the single `structure` argument nests exactly; each row is priced and inverted by the same two calls.

model	covariance grammar	source
Luce / softmax	independent, Gumbel base, $D = \pi^2/6$	exact identity
Thurstone–Mosteller	independent, normal base	Cotton (2021)
factor multinomial probit	$VV' + \mathrm{diag}(D)$	companion note
teams, conferences	blocks: rank-r effect per cluster	§5
market plus sectors	nested: blocks $\times$ global factor	§5
hierarchies	tree: uniform shared effects	§6
dense Σ, estimable case	fitted grammar + residual factors	§8

Scale motivates second. The next table reports measured wall clock for all n probabilities (Rust kernels, one laptop) against GHK's cost law, measured as $n^{2.8}$ up to $n = 10^3$ at $R = 10^4$ draws and extrapolated beyond. At these budgets GHK's total variation error is near 10^{-2}; the lattice error is near 10^{-8}.

n	lattice, measured	GHK at 10^4 draws, cost law	multiplier
10^4	0.18 s	3.7 days	2×10^6
10^5	2.7 s	6.5 years	8×10^7
10^6	29 s	4,100 years	4×10^9

Sections 2 places the method among the alternatives. Sections 3–6 describe the algorithm: the independent lattice race, then the factor, block, nested and tree extensions, each with its recursion written out. Section 7 gives the Jacobian and the inversion. Section 8 reduces dense covariance to the grammar family and reports accuracy per named random-correlation ensemble. Section 9 contains the benchmarks.

2 The alternatives

Deterministic quadrature. Genz (1992) transformed the orthant integral to the unit cube and integrated by adaptive and quasi-Monte Carlo rules; the `mvtnorm` implementation is the standard in R. Accuracy is not the issue: in the benchmark below it agrees with the lattice to the reference's own noise. The shape is the issue. It treats one probability at a time, so the all-n vector at $n = 30$ costs three hundred times the lattice, and the gap widens with n.

Frequency simulation. The crude frequency estimator of Lerman and Manski (1981) counts argmax draws. It is unbiased for the whole vector at once but not smooth in parameters, and its per-entry relative error explodes for small p_i. Measured below: at wall clock matched to the lattice it carries twelve to twenty times the total-variation error, records zero counts on tail alternatives at $n = 30$, and at ten times the budget remains a factor of two to four behind.

GHK. Defined above. Two properties matter for comparison: it is smooth in parameters, which frequency counting is not, and it prices one alternative per run. Stern (1992) gave a related decomposition with an analytic leading term.

Bayesian data augmentation. Albert and Chib (1993) and McCulloch and Rossi (1994) sidestep the integral entirely by Gibbs sampling the latent utilities. The cost moves into the Markov chain, and the likelihood surface itself is never available, which forecloses everything that needs it. No pricing benchmark is possible for the same reason: the method never produces the probabilities.

Analytic approximations. Clark (1961) matched moments of pairwise maxima; Mendell and Elston (1974) and Solow (1990) approximate by sequential conditioning on first and second moments. These are fast, sometimes startlingly accurate, and carry no error control, and the benchmark below measures all three properties at once: Mendell–Elston prices the $n = 10$ vector in two milliseconds within 2×10^{-3} of the reference, then drifts to 3×10^{-2} at $n = 30$ with tail probabilities off by a factor of nearly four, and nothing in the output announces the change. The Clark-type recursion behaves alike. Expectation propagation is the modern member of this family and inherits its profile; we benchmark the family through its classical members.

Minimax tilting. Botev (2017) computes single Gaussian rectangle probabilities with bounded relative error deep into the tails, in dimensions up to roughly a thousand. It is the accuracy reference, and we use it as the referee for our own tail claims; it prices one probability per run.

Every method above computes one probability at a time. Estimation wants all n and their derivatives at every parameter value. Sharing the field across alternatives is the algorithmic point of this paper, and it is what makes the n-fold and the derivative costs disappear.

3 The lattice race

Fix a min-wins convention: performances are times, the lowest wins, and $U_i = -X_i$ translates to the utility convention. Let $X_i = \mu_i + \sigma_i \epsilon_i$ with ϵ_i i.i.d. from a standardized base density (normal, Gumbel, or any density supplied as code), and write f_i, S_i for the density and survival of X_i.

The algorithm evaluates (2) on a lattice $x_1 < \cdots < x_L$:

1. compute $\log S_j(x_\ell)$ and $\log f_j(x_\ell)$ for all j, ℓ;
2. accumulate the field $L(x_\ell) = \sum_j \log S_j(x_\ell)$;
3. for each i, form the integrand $f_i(x_\ell)\, e^{L(x_\ell)-\log S_i(x_\ell)}$, which is contestant i's density times everyone else's survival, with i divided out of the shared field in log space;
4. sum times the lattice spacing, and normalize the vector.

The cost is $O(nL)$: the field is built once, not once per contestant. Log-space arithmetic keeps hopeless contestants finite (their $e^{L-\log S_i}$ underflows gracefully rather than dividing zero by zero), and the exponent is clipped at -745.

The lattice itself is placed on the winner's bulk, not the ability span. A hopeless contestant wins only by running a winner-class time, so only the region where the winning time lives matters. Bisect the conservative winner c.d.f. envelope $G(x) = 1-\prod_j S_j(x)$ to $[G^{-1}(\delta), G^{-1}(1-\delta)]$ and pad two standard deviations so that no base's tails are clipped. Measured: 33 points on this window reach accuracy that a 500-point ability-span lattice cannot, because the span window's own truncation floors near 6×10^{-11}; on fields with many hopeless contestants the bulk window is one hundred times more accurate at equal budget.

Two byproducts are free. The same pass yields removal counterfactuals, the probability j wins after deleting i, by dividing a second contestant out of the field; and photo-finish tie densities, which §7 identifies with the off-diagonal Jacobian.

4 Factor covariance

Let $X = \mu + Vf + \mathrm{diag}(\sqrt{D})\,\epsilon$ with $f \sim N(0, I_k)$ independent of ϵ, so $\Sigma = VV' + \mathrm{diag}(D)$. Conditional on f the performances are independent with means $\mu + Vf$, and

$$p \;=\; \sum_q w_q\, p^{\mathrm{ind}}(\mu + V f_q), \tag{3}$$

a Q-node quadrature over runs of the independent algorithm, cost $O(nLQ)$. In English: average the independent race over the shared luck. Gauss–Hermite nodes serve $k \le 2$; scrambled Sobol serves $k \ge 3$.

One default matters. When $D_i \ll \|V_i\|^2$ the conditional race is nearly deterministic and the integrand over f is nearly a step function; a fixed 15-node rule then silently loses up to five percent of total variation, and the loss is invisible to lattice refinement because it is points-invariant. Randomized invariant testing against a QMC referee found this. The default order now scales with the sharpness $\max_i \|V_i\|/\sqrt{D_i}$, capped by rank.

The same pass returns own-ability slopes $\partial p_i/\partial \mu_i$ at no extra cost, which §7 uses as the Newton preconditioner.

5 Block and nested covariance

Assign contestant i to cluster $c(i)$ with loading $v_i \in \mathbb{R}^r$ and let

$$X_i = \mu_i + v_i' a_{c(i)} + \sqrt{D_i}\,\epsilon_i, \qquad a_c \text{ i.i.d. } N(0, I_r), \tag{4}$$

a block-diagonal rank-r-plus-diagonal covariance: teams, conferences, sibling products. Two independences do the work. Conditional on its own cluster's effect a contestant is independent of its teammates; and distinct clusters are independent outright. So the field factorizes across clusters,

$$\prod_j S_j(x) \;=\; \prod_c G_c(x), \qquad G_c(x) \;=\; \mathbb{E}_a \prod_{j \in c} S_j(x \mid a) \;=\; \sum_q w_q\, e^{S_c(x, a_q)}, \tag{5}$$

with $S_c(x, a) = \sum_{j \in c} \log S_j(x \mid a)$ the cluster's conditional log-survival. In English: $G_c(x)$ is the probability that every member of cluster c survives past x, with the cluster's shared luck integrated out.

Contestant i then wins against a cavity field:

$$p_i \;=\; \int h_i(x) \prod_{c \neq c(i)} G_c(x)\, dx, \qquad h_i(x) \;=\; \sum_q w_q\, f_i(x \mid a_q)\, e^{S_{c(i)}(x, a_q) - \log S_i(x \mid a_q)}. \tag{6}$$

The term h_i couples i's win density to its own teammates within the shared draw of $a_{c(i)}$; the cavity product supplies the rest of the field. The cost is $O(nLQ_a)$ whatever the number of clusters, because the per-cluster factors are accumulated once. Rank two is special: the conditional integrand is smooth and a tensor Gauss–Hermite rule beats scrambled Sobol at equal nodes by a wide measured margin.

Nested. Add one global factor with couplings g_i and a dial $\gamma \in [0, 1]$: conditional on the global draw f_q the model is a block race at shifted abilities, so $p = \sum_q w_q\, p_{\text{block}}(\mu + \gamma g f_q)$, a finite mixture of block races. The dial interpolates Schur-complementary beliefs from isolated clusters to full coupling.

6 Tree covariance

Let a rooted tree have the leaf clusters at its bottom and internal nodes t above, and give node t a scalar effect $b_t \sim N(0, 1)$ applied with uniform strength λ_t to every leaf beneath it:

$$X_i = \mu_i + v_i\, a_{c(i)} + \sum_{t \,\in\, \text{anc}(c(i))} \lambda_t b_t + \sqrt{D_i}\,\epsilon_i. \tag{7}$$

The implied covariance is hierarchical: two contestants share the variance of exactly their common ancestors. The uniform strength is the crucial restriction, because a shared effect that hits every subtree member equally moves the whole subtree's field by a translation of its argument. Messages therefore remain functions of one variable. Per-leaf loadings on internal effects would force two-dimensional message tables.

The algorithm is two passes over the same lattice.

Upward. Each leaf cluster carries its block factor $G_c(x) = \sum_q w_q e^{S_c(x,a_q)}$ from (5). Each internal node, visited deepest first, combines its children under its own effect:

$$G_t(x) \;=\; \sum_q w_q \prod_{c\,\in\,\mathrm{children}(t)} G_c(x + \lambda_t a_q)\,. \tag{8}$$

In English: $G_t(x)$ is the probability that every leaf below t survives past x, with t's effect integrated by quadrature and each child's message shifted because the effect moves that child's whole subtree at once. Shifted evaluations are linear interpolations on the lattice.

Downward. The root's cavity is one. Each node, visited shallowest first, receives the field of everything outside its subtree:

$$R_t(x) \;=\; \Big[\sum_q w_q\, R_{\mathrm{parent}(t)}\big(x - \lambda_{\mathrm{parent}(t)} a_q\big)\Big] \prod_{s\,\in\,\mathrm{siblings}(t)} G_s(x). \tag{9}$$

Contestant i then wins against its own leaf cavity, $p_i = \int h_i(x)\, R_{c(i)}(x)\, dx$ with h_i as in (6). The cost is $O(nLQ)$ plus $O(L\,Q)$ per tree node.

A depth-one tree with zero strengths reproduces the block race to machine precision, which is tested. Against a 2^{22}-path common-random-numbers referee the recursion sits at the simulation noise floor on every resolvable entry at depths one through four (root-mean-square z between 0.75 and 1.1). Entries below the referee's resolution agree with minimax tilting (Botev, 2017) to 0.3 percent relative at probabilities down to 7×10^{-8}, which is within the tilting estimate's own error. Tail accuracy is a property of the lattice, not of node placement, so the grammar kernels price the tails that simulation cannot see.

7 Jacobians, tie densities, inversion

Proposition 1 (Off-diagonals are photo-finish densities)**.** *For the race* (2) *and* $j \neq i$*,*

$$\frac{\partial p_i}{\partial \mu_j} \;=\; \int f_i(x)\, f_j(x) \prod_{k \neq i,j} S_k(x)\, dx \;\geq\; 0, \tag{10}$$

and each row of the Jacobian sums to zero.

The zero row sum says a common shift moves nothing. The off-diagonal is the density of a dead heat between i and j ahead of the field, so the Jacobian is a byproduct of the same field pass: per quadrature node it is a Gram matrix over the lattice. The stable pair factorization is $[f_i e^{L - \log S_i}] \times [f_j / S_j]$, a bounded density times a polynomially growing hazard; the symmetric square-root split overflows where survivals vanish.

The block Jacobian is exact, with the within-cluster term computed under the cavity; the nested Jacobian is the mixture of block Jacobians; the tree Jacobian is exact within a cluster and Gram-approximate across clusters, which suffices for Newton because residuals always use the exact forward map. Feasibility-critical callers fall back to finite differences.

Inversion recovers mean-zero μ from a target p: an adaptive damped fixed point on the log residual carries the iterate into Newton's basin, then Newton steps on $\log p(\mu) - \log p^\star$ solve to 10^{-8}; sub-resolution targets are floored and reported as bounds. Ignoring known structure at inversion time is not a small error: inverting block-generated probabilities under an assumed independent model mislocates abilities by up to a fifth of the field's spread.

8 Dense covariance

Fit the grammar family to a dense Σ: rank-k global factors, then hierarchically seriated blocks with one rank-one effect each fitted to the within-block residual, alternated to convergence with the diagonal excluded from every fit; then promote the top m eigencomponents of the remaining residual to additional factor columns. What results is a factor-plus-blocks race priced in one deterministic call. At $n = 2000$ the call takes five seconds against thirty-six for a 10^6-draw simulation, agrees with it at its noise floor, and prices the 82 percent of tail alternatives for which the simulation records zero wins.

A random covariance matrix means nothing until the measure is named, so accuracy is reported per named ensemble of the `randomcov` library. Median absolute error is $1\text{–}4 \times 10^{-5}$ under twelve of fifteen measures: factor with sparse links, sparse-precision graphical, Wishart, Marchenko–Pastur and spiked spectra, LKJ, onion, vine, hierarchical, block equicorrelation, AR(1) and elliptope walks. It is $1\text{–}4 \times 10^{-4}$ under the dense-strong ensembles, and it fails, at 4×10^{-3}, under Matérn and RBF kernel fields. Kernel covariance is locality, not spectrum: such fields are approximately Markov (Lindgren et al., 2011) and their polynomially decaying spectra offer a low-rank fit nothing to hold. The right algorithm there is sequential conditioning with cheap sparse-precision draws, and a production front door should dispatch on the structure it detects.

9 Benchmarks

Against the field. All-n choice probabilities under the rank-2 factor covariance family of Section 2, against a 2^{20}-point QMC reference. Tail is the maximum absolute log-error over reference probabilities in $[10^{-4}, 10^{-3}]$; the $n = 10$ draw has none. Frequency simulation is run at wall clock matched to the lattice and at ten times that budget; "zeros" counts tail alternatives that received no draws.

	$n = 10$		$n = 30$		
method	time	TV	time	TV	tail
lattice race	2.5 ms	1.8×10^{-4}	4.7 ms	8.3×10^{-4}	0.07
Genz, full vector	46 ms	1.8×10^{-4}	1.40 s	8.3×10^{-4}	0.07
frequency, matched time	—	3.8×10^{-3}	—	1.0×10^{-2}	0.45, 2 zeros
frequency, 10× time	—	9.2×10^{-4}	—	2.0×10^{-3}	0.21
Mendell–Elston	2.0 ms	1.8×10^{-3}	18 ms	3.2×10^{-2}	1.30
Clark-type	1.4 ms	1.1×10^{-2}	17 ms	1.6×10^{-2}	1.33

Genz and the lattice agree to the reference's noise and to each other; the 300× cost ratio at $n = 30$ is the one-at-a-time shape, not the integrator. The sequential-conditioning family is the only entry whose error grows silently: its tail entries are off by $e^{1.3} \approx 3.7$ with nothing in the output to say so. The race's tail figure is the reference's own noise; lattice self-convergence places its error near 10^{-8}.

Against GHK. All-n choice probabilities under a rank-2 factor covariance, against a 2^{20}-point QMC reference:

	lattice race		GHK, $R = 10^3$		GHK, $R = 10^4$	
n	time	TV err	time	TV err	time	TV err
10	2.3 ms	$\leq 1.8 \times 10^{-4}$	1.0 ms	8.4×10^{-3}	8 ms	2.1×10^{-3}
50	7.0 ms	$\leq 1.7 \times 10^{-3}$	21 ms	4.2×10^{-2}	222 ms	1.2×10^{-2}
200	25 ms	$\leq 2.4 \times 10^{-3}$	436 ms	3.2×10^{-2}	5.0 s	1.2×10^{-2}

The lattice bounds are the reference's own noise; self-convergence places the lattice error near 10^{-8}. GHK error contracts as $O(R^{-1/2})$ and its measured cost grows as $n^{2.8}$ per probability vector. The lattice is deterministic, supplies the exact Jacobian that maximum simulated likelihood lacks, and prices the tails. Identification restricts multinomial probit to structured covariance in any case, so the structured case is the estimable case, and there we consider GHK replaced. GHK retains general rectangle probabilities and locality-structured covariance, and Botev (2017) remains the reference for single extreme tail probabilities.

Estimation. The pricing advantage transfers to estimation. Simulate $T = 50{,}000$ choices among $n = 30$ alternatives under a known rank-2 factor covariance and estimate the mean-zero utilities by maximum likelihood, exact probabilities and analytic score against GHK probabilities with common random numbers and finite-difference gradients, eight replications:

estimator	RMSE over identified alternatives	median fit time
exact gradient MLE	0.0187	4.9 s
MSL, $R = 100$	0.0305	4.4 s
MSL, $R = 1000$	0.0198	40.7 s

At equal fitting time the simulated estimator pays a 63 percent error premium; at eight times the budget it remains behind. The exact estimator sits at the statistical floor.

Scale. The Rust kernels are input-bound: ten million contestants in 64 seconds under the block grammar with a hybrid in-memory and streaming field, and 232 times the speed of the interpreted classic calibration.

Parity. One vector file embeds the inputs and outputs of 22 scenarios spanning every claim above. Python, R, JavaScript and Rust replay them: twenty at machine precision, the optimizer-mediated scenarios to 10^{-6}.

Remark 1 (A portfolio identity)**.** *Long-only portfolio weights are race probabilities, and the covariance implicitly assumed by hierarchical risk parity (López de Prado, 2016) is exactly a tree race: give internal node t strength $\lambda_t^2 = \rho_t - \rho_{\mathrm{parent}(t)}$ with $\rho_t = \max(1 - 2h_t^2, 0)$ for merge height h_t, and leaf i idiosyncratic variance $1 - \rho_{\mathrm{first}(i)}$; the implied correlation equals the floored cophenetic matrix. The floor is forced, because uniform shared effects cannot represent negative dependence, and omitting it silently inflates other implied correlations above one. The identity turns dendrogram-consistent concentration limits into contest inversions, but portfolio applications are not this paper's subject.*

10 Availability

`pip install winning` is pure Python; `winning[fast]` adds the Rust kernels as one abi3 wheel per platform. A base-R package mirrors the full interface, and a zero-dependency browser port drives a live demonstration at `winning.microprediction.org`.

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